4 **(a)** Define *electric field strength*.

Electric field strength at a point is defined as the **electric force per unit positive charge** exerted on a **small test charge** placed at that point. [1]

(b) An oil drop of mass 1.0×10^{-15} kg and carrying a charge of 1.6×10^{-19} C is placed in the middle of two horizontal parallel plates.

The separation between the plates is 40 mm and the the potential difference between the plates is 8.8 kV. The oil drop is seen to be attracted towards the upper plate because of its electric charge.

(i) Calculate the electric force experience by the oil drop.

Electric force = qE

$= q\frac{V}{d}$ [1]

$= (1.6\times10^{-19})\frac{8.8\times10^{3}}{40\times10^{-3}}$

$= 3.52\times10^{-14}$ N [1]

force = ... N [2]

(ii) Initially the oil drop is 20 mm from either plate and is at rest. Ignoring air resistance, use energy considerations to calculate its speed when it reaches the upper plate.

By Conservation of Energy,
Gain in kinetic energy + Gain in GPE = Loss in electrical potential energy

$\frac{1}{2}mv^2 - 0 = q\Delta V + mg\Delta h$ [1]

$\frac{1}{2}(1.0\times10^{-15})v^2 = (1.6\times10^{-19})\left(\frac{8.8\times10^{3}}{2}\right) + (1.0\times10^{-15})\times9.81\times(20\times10^{-3})$

$v = 1.01\text{ m s}^{-1}$ [1]

speed = ... m s^{-1} [2]

[Total: 5]